

2023 NATIONAL YOUTH RISK BEHAVIOR SURVEY

Co-Occurring Adverse Experiences & Protective Factors in Adolescent Suicidality

A survey-weighted analysis of self-reported
adversity, protective behaviours, and suicide-risk
outcomes among U.S. high school students





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01

Introduction

An introduction to our project,
key aims and objectives

Background



2021



Emergency

American Academy of Child and Adolescent Psychiatry declared a "national emergency in child and adolescent mental health" in the U.S.

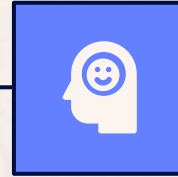
2022



Hearing

The U.S. Senate HELP Committee held a formal hearing, *"Why Are So Many American Youth in a Mental Health Crisis?"*

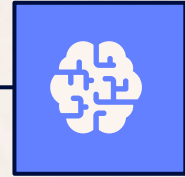
2023



Evidence

CDC's national survey quantifies the scale of the crisis among U.S. high schoolers

2026



Investigation

Our project examines which adverse and protective experiences co-occur with this risk



Background

Suicide is the second leading cause of death among Americans aged 10-34

Over the past decade, rates of serious suicidal ideation have risen

Background

1 in 5

seriously considered attempting
suicide in the past 12 months

39.7%

of students reported persistent
sadness or hopelessness
in the past 12 months

9.5%

attempted suicide one or more times
in the past 12 months

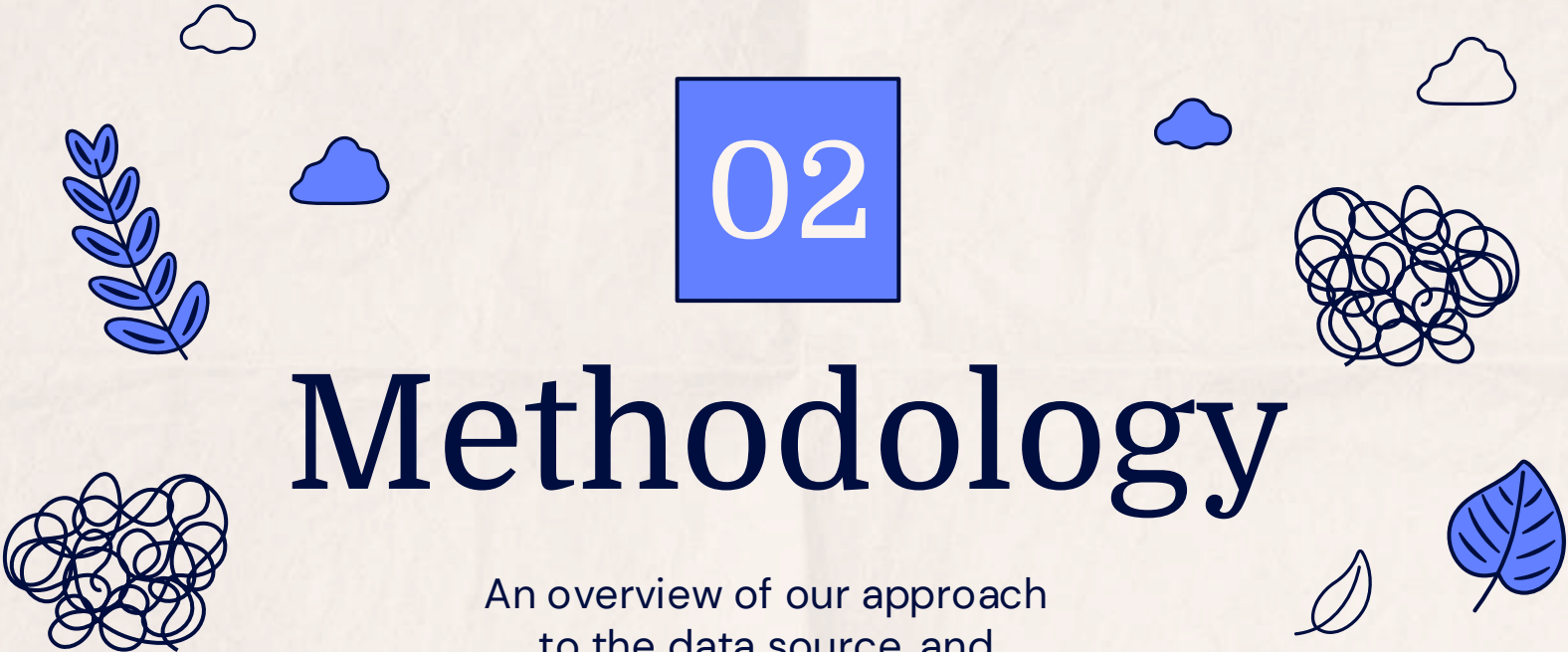




Background

Suicide risk rarely comes from one thing.
This gap motivates our research question:

how do adverse and protective experiences co-occur with adolescent suicidality, and what does that mean for where prevention efforts should focus?



02

Methodology

An overview of our approach
to the data source, and
methods of collaborative

Study design: 2023 National YRBS



What is it?

A nationally representative, biennial CDC survey of U.S. high school students (grades 9-12)

Anonymous, self-report questionnaire: every value in this dataset is what a student said, not something independently verified

20,103 usable student responses in the 2023 cycle

Three-stage stratified cluster design:
strata → primary sampling units (PSUs, roughly clusters of schools) → classrooms



Why does this design matter?

Weight

Each respondent stands in for a number of real students nationally (not every student in the survey represents just one person)

stratum / psu

Students in the same school answer more alike than random students elsewhere.

Ignoring this understates uncertainty.

Project structure

	Domain	Details	Variables
Jacinta	Demographics		Q1, Q2, Q3, raceeth
Aliya	Protective factors		QN85, QN87, QN99, QN103, QN104
Destiny	Adverse factors	Sexual dating and violence	QN19, QN20, QN21, QN88
Albert	Adverse factors	School and community	QN22, QN23, QN24, QN25, QN86
Terence	Adverse factors	Household dysfunction	QN89, QN90, QN91, QN100, QN101, QN102
Francis	Risk behaviours	Violence and mental distress	QN12, QN13, QN16, QN26, QN84
Bernard	Risk behaviours	Substance use	QN43, QN48, qntb4, qnillict, qnowt, qnobese

Study design: How we analysed it



Accounting for clustering

YRBS surveys students in clusters by school, not individually at random.

So, treating each row as independent understates true uncertainty in our own comparisons.



Design-based estimation

Every prevalence estimate uses design-based estimation (Taylor series linearisation), incorporating stratum, PSU, and survey weight.

The same method the CDC uses in its own official YRBS analyses.



Independent validation

Before trusting our own results, we independently reproduced CDC's published national figures from the raw data, and matched them.

Reproducibility via GitHub

Everything lives in one place

All scripts, data, documentation, and outputs are tracked in a single public repository to see the full history of the project

Every change is logged and attributed

21 commits across the project, each with a description of what changed and who changed it (from variable selection to charts)

Structured for reproducibility

Separate folders for data, scripts, output, markdown, and documents the pipeline from raw data to final figures is traceable end to end.

Analysis can be re-run from scratch

Anyone can rebuild our results because the R scripts and outputs are version-controlled together

Commits

History for BIP-Technical-Challenge-Collaboration / documents on [main](#)

Commits on Aug 17, 2026

Risk Behaviours

 FrancisAsane authored 3 days ago

Add files via upload

 ali-pixel545 authored 3 days ago

Commits on Aug 16, 2026

Add files via upload

 ajsastheticjourneys authored 4 days ago

Commits on Aug 14, 2026

Updated to include QNILLICIT variable

 bernkyl authored last week

Add files via upload

 Terence03-art authored last week

Commits on Aug 12, 2026

Add files via upload

 ajsastheticjourneys authored last week

Commits on Aug 9, 2026

Add files via upload

 cakemaster1017 authored 2 weeks ago

Add files via upload

 ali-pixel545 authored 2 weeks ago

Commits on Aug 5, 2026

Variable Research

 bernkyl authored 2 weeks ago

Collaboration via Google Drive & Teams

Shared with me > BIP Technical Challeng... > Recordings ▾

◆ Ask Gemini | Type ▾ People ▾ Modified ▾ Source ▾

Name ↓	Owner	Date modified
 Wednesday Meeting Team 04.-20260729_141217... 	 chukwuedojacinta	29 Jul
 Team 04 Meeting.-20260802_192840-Meeting R... 	 chukwuedojacinta	7 Aug
 Team 04 Meeting.-20260726_190616-Meeting Re... 	 chukwuedojacinta	26 Jul





03

Key findings

A breakdown of key results
across the four suicide outcomes

Destiny's variables

Variable	Measures	Missing
QN19	Ever physically forced to have sexual intercourse	13.9%
QN20	Forced to do sexual things, past 12 months	21.6%
QN21	Sexual dating violence (among students who dated)	33.1%
QN88	Sexual contact from someone 5+ years older	31.2%

2.8x

higher lifetime suicide risk among sexual violence survivors vs. non-survivors (Mondin et al., Brazil, n=1,560)

27.3% vs 9.4%

suicidality prevalence in sexually assaulted vs. unassaulted subsamples (Dworkin et al., meta-analysis)

Na et al.'s review of meta-analyses names childhood sexual assault among the strongest identified risk factors for suicide attempt.

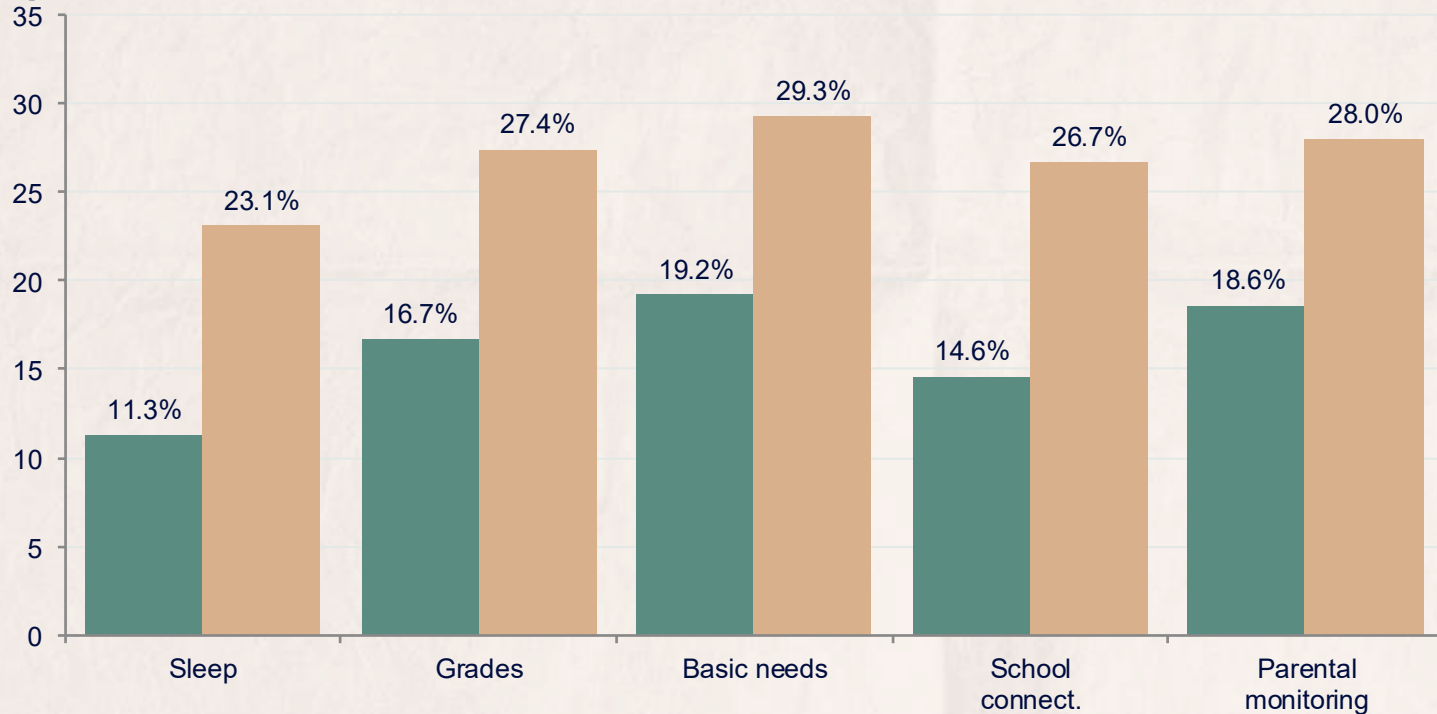
Terence's variables

Variable	Description	Missing
QN89 / QN90	Verbal or physical abuse by a household adult	OR 1.5-3x (verbal); 2-3x (physical) higher odds of suicidal ideation/attempts (Liu et al. 2017; Miller et al. 2013; Norman et al. 2012)
QN91	Domestic violence witnessed in the home	Substantially increased Lifetime suicide-attempt risk (Felitti et al. 1998, original ACE Study; Hughes et al. 2017)
QN100	Parental/guardian substance use problem	2-3x increase in odds of lifetime suicide attempt (Felitti et al. 1998); effect persists after controlling for sociodemographics (Hughes et al. 2017)
QN101	Parental/guardian mental illness or suicidality	Roughly doubled odds of lifetime attempt (Felitti et al. 1998); one of the strongest household-level predictors in prior YRBS cycles (Baiden et al. 2021)
QN102	Separation from a parent due to incarceration	Significantly elevated mental health problems, including suicidal thoughts and self-harm (Murray et al. 2012)

All classified as adverse experiences, for household dysfunction

Aliya's variables

% who seriously considered suicide (QN27), by protective factor



**HOLDS ACROSS
ALL 20 COMPARISONS**

Every 1 of the 5 protective factors showed lower prevalence on **every one** of the 4 suicidality outcomes:

1. Considered
2. Planned
3. Attempted
4. Injured

Protective factors

■ Has the protective factor ■ Does not



Aliya's variables (protective factors)



Factor	CDC adjusted PR (2023 YRBS)	Our ratio (unadjusted)	Agreement
Sleep (QN85)	0.53 - 0.67	0.48 - 0.49	Close match
Basic needs met (QN99)	0.41 - 0.80	0.23 - 0.66	Close match
Parental monitoring (QN104)	0.51 - 0.74	0.34 - 0.66	Close match
School connectedness (QN103)	0.63 - 0.70	0.52 - 0.56	Close match
Grades (QN87)	Not tested by CDC	0.34 - 0.66	Consistent with outside literature

A ratio below 1.0 = lower risk in the protective-factor group. 0.53 $\approx$ roughly half the risk.

Sleep, basic needs, parental monitoring and school connectedness are the exact protective factors CDC selected for its own 2023 YRBS analysis: and our unadjusted ratios sit close to their adjusted ones.

Grades isn't one of CDC's six protective factors, so we drew on outside literature instead — directionally consistent, but the least consistently supported of the five across subgroups in the wider research.

Francis' variables

Variable	Question	Findings
QN12	Carried a weapon (gun, knife, or club) on school property, past 30 days	Onoh et al. (2008): weapon carrying independently associated with suicidal ideation (OR=1.64), depression (OR=1.44), and physical fighting (OR=2.02)
QN13	Carried a gun, past 12 months (excluding hunting/sport)	Kagawa et al. (2019): 2.4% prevalence (vs. 3.5% here); 65% of gun carriers met criteria for a psychiatric disorder; linked to conduct disorder (APR=1.88), drug use disorder (APR=1.91), specific phobias (APR=1.54). Baiden et al. (2024): significantly associated with sad/hopeless (OR=1.61)
QN16	Number of physical fights, past 12 months	Macková et al. (2021): family-related adversity significantly predicted more fighting ($p<.001$), mediated by hopelessness (indirect effect $ab=0.04$, 95% CI: 0.02–0.05) — a plausible mechanism linking sadness/hopelessness to fighting
QN26	Felt sad or hopeless almost every day for 2+ weeks, past 12 months	Baiden et al. (2024): significant bivariate association with gun carrying (OR=1.61, $p<.001$), not significant after full adjustment (AOR=0.77, $p=.188$)
QN84	Self-rated poor mental health, past 30 days	Baiden et al. (2024): significant bivariate association with gun carrying (OR=1.32, $p=.027$), not significant after full adjustment (AOR=0.99, $p=.961$)

Bernard's variables

Variable	Question	Findings
QN43	Had 4+ (female) or 5+ (male) drinks in a row, at least 1 day in the past 30 days	Onoh et al. (2008): weapon carrying independently associated with suicidal ideation (OR=1.64), depression (OR=1.44), and physical fighting (OR=2.02)
QN48	Used marijuana, past 30 days	Evidence on marijuana's link to suicidal ideation is mixed: some studies report a 50-65% increase in risk during adolescence, while others find significance only among heavy users.
QNTB4	Number of physical fights, past 12 months	A 28-country adolescent study found a positive direct relationship between cigarette smoking and suicide attempts via moderated mediation analysis, motivating a check of whether this pattern holds in YRBS data.
QNILLICIT	Used cocaine, inhalants, heroin, methamphetamines, ecstasy, or hallucinogens 1+ times, lifetime	Tunagur's research highlights a correlation between high suicide risk and frequent drug use. Because this variable captures only whether use occurred (not frequency), it's unclear whether that pattern will hold here
QNOWT, QNOBESE	Self-rated poor mental health, past 30 days	In adults, overweight/obesity has been linked to suicidal ideation, particularly in female professionals under 35. Young adults specifically, studies show a strong correlation with suicidal ideation, particularly among young men



Albert's variables



QN22 (physical dating violence) has a weighted prevalence of 10.4%. Baiden, Mengo and Small (2019), using 2015 national YRBS data, found students with a history of physical teen dating violence were 1.92x more likely to report suicidal ideation, 1.67x more likely to have made a plan, and 2.42x more likely to have attempted suicide than peers without this history.



QN23 (racial/ethnic discrimination in school) captures a distinct adversity dimension rooted in identity-based discrimination rather than interpersonal violence. Baiden and colleagues (2022) found perceived racial discrimination in school was significantly associated with suicidal ideation, planning, and attempts among racial and ethnic minority adolescents.

QN24 and QN25 (school bullying and electronic bullying) analysed by Messias, Kindrick and Castro (2014) found bullying victimisation — school, electronic, or both — strongly associated with sadness and suicidality, with the highest odds among students experiencing both forms. A meta-analysis by Holt and colleagues (2015) confirmed bullying involvement is consistently associated with suicidal ideation and attempts across dozens of studies.

QN86 (unstable housing) is unlike the other four variables in this group in that it reflects a structural/socioeconomic circumstance rather than an interpersonal event. Added to the national YRBS for the first time in 2021, CDC's own analysis (McKinnon et al., 2023) found that although only 2.7% of US high school students experienced unstable housing nationally, these students reported substantially higher rates of suicidal ideation and attempts than stably housed peers.

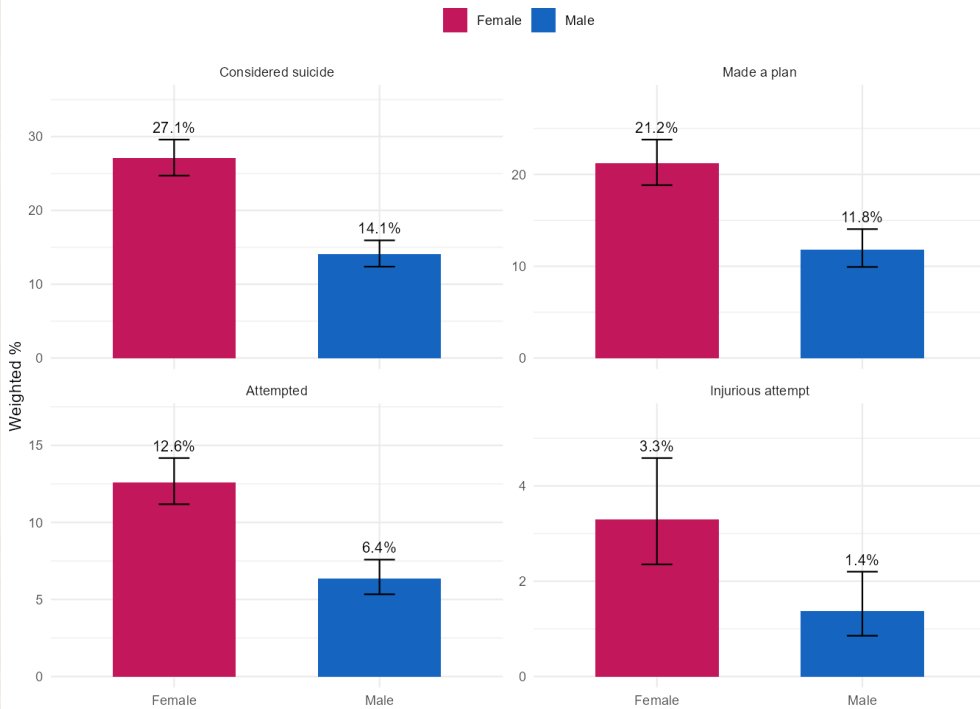


Jacinta's variables



Suicidality Outcomes by Sex

Weighted % with 95% CI, all four outcomes



Overall

Sex shows the strongest and most consistent demographic pattern in this dataset.

Female students report significantly higher prevalence than male students across all four outcomes

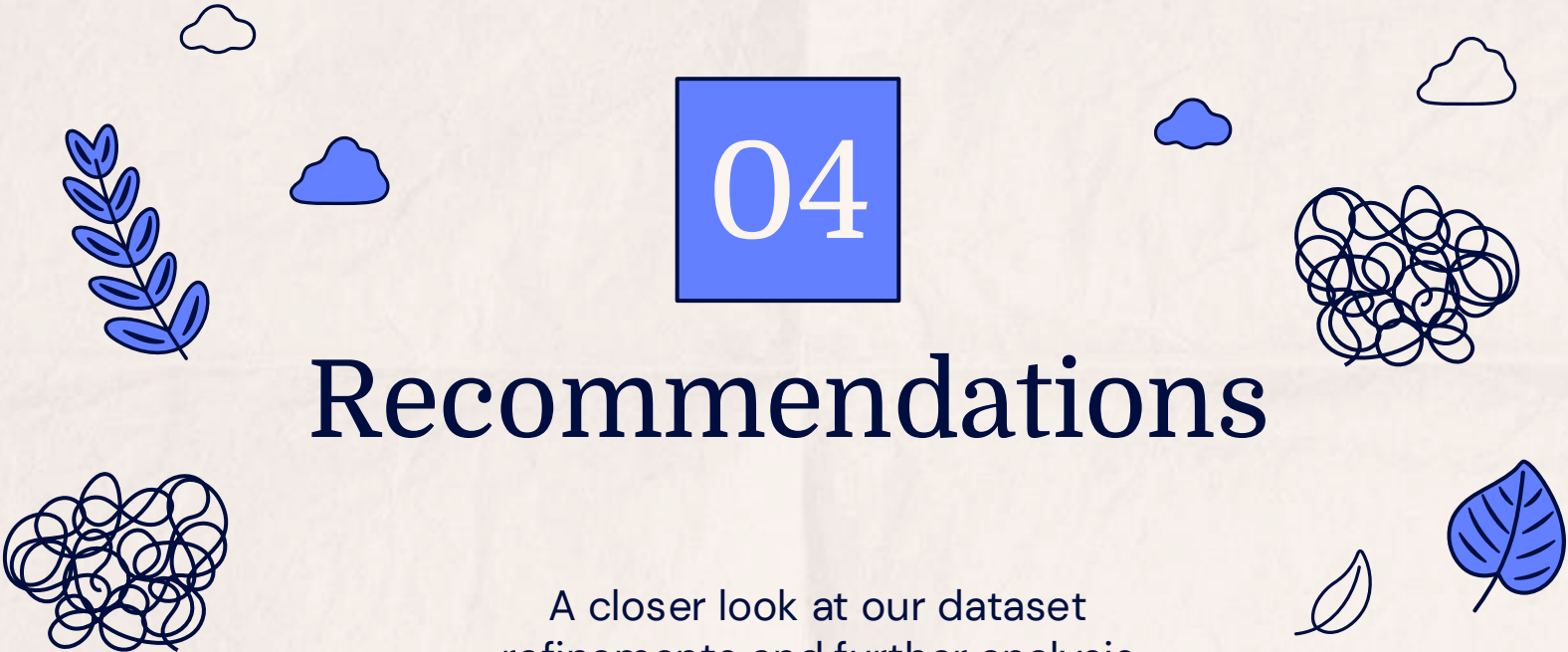


Composition

- White 48.1%
- Multi-Hispanic/Latino 19.8%
- Black 13.3%
- Hispanic 7.7%
- Multi-non-Hispanic 6.1%
- Asian 4.3%

Key findings

Race/ethnicity's relationship with suicidality is inconsistent across the four outcomes — statistically significant for considered suicide ($p=0.003$) and attempted suicide ($p=0.008$), but not significant for made a plan ($p=0.100$) or injurious attempt ($p=0.082$).



04

Recommendations

A closer look at our dataset
refinements and further analysis



Limitations



Direction and causality

This is cross-sectional, self-report data. We can't establish that a protective factor came before (or caused a change in) an outcome, only that they co-occur



Recall & reference periods

Different questions ask about “during your life,” “past 12 months,” and “past 30 days”. These inconsistent windows can affect recall and comparability



Overlapping factors

Protective factors are interrelated (i.e., a well-monitored student may also sleep more), so this analysis can't isolate each factor's independent contribution



Unmeasured cofounders

Household socioeconomic status is known to shape both protective factors and mental health outcomes, however isn't captured in the YRBS



05

Conclusions

A summary of our conclusions,
limitations, and next steps

Two sides of the same picture



Adverse experiences

- Sexual & dating violence,
- household dysfunction, school & community adversity, violence-adjacent behaviours each individually linked to higher suicide-risk indicators in prior research

↑ higher prevalence



Protective factors

- Sleep, grades, basic needs met, school connectedness, parental monitoring, each associated with substantially lower prevalence of every suicidality outcome measured.

↓ lower prevalence

Thanks!

Does anyone have any questions?



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